FLAT PULLBACK COMMUTES WITH PROPER PUSHFORWARD FOR A FIBRE PRODUCT SQUARE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Properness and flatness of relative dimension are preserved by base change. Morphisms, labels morphisms-lemma-base-change-proper (proof line 2422), morphisms-lemma-base-change-relative-dimension-d and morphisms-lemma-base-change-flat (proof lines 2424-2425). Gives the first two assertions of the lemma, that the base changed morphisms are proper and flat of the same relative dimension.
- The cycle class of a closed subscheme, flat pullback and proper pushforward of cycle classes, and cohomology and base change. Same chapter chow.tex, labels lemma-cycle-closed-coherent (proof line 2429, claimed at a withdrawn revision), lemma-cycle-push-sheaf (proof line 2430) and lemma-pullback-coherent (proof line 2431); Cohomology of Schemes, label coherent-lemma-flat-base-change-cohomology (proof line 2435). Reduces the equality of cycles to an isomorphism of the two pushforwards of the structure sheaf of an integral component, which is the flat base change theorem.

Source reference. Original lemma 02KH: sources/stacks-chow/tags.txt, source lines 3372–3372. Navigation pointer for \ref{coherent-lemma-flat-base-change-cohomology} used inside the counted span (line 2435); the argument that consumes it is retained in the span.

Source reference. Original lemma 02QY: sources/stacks-chow/pinned-original-source.tex, source lines 1776–1776. Navigation pointer for \ref{lemma-cycle-closed-coherent}

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used inside the counted span (line 2429); the argument that consumes it is retained in the span.

Source reference. Original lemma 02R6: sources/stacks-chow/pinned-original-source.tex, source lines 1992–1992. Navigation pointer for \ref{lemma-cycle-push-sheaf} used inside the counted span (line 2430); the argument that consumes it is retained in the span.

Source reference. Original lemma 02RE: sources/stacks-chow/pinned-original-source.tex, source lines 2331–2331. Navigation pointer for \ref{lemma-pullback-coherent} used inside the counted span (line 2431); the argument that consumes it is retained in the span.

Source reference. Original lemma 01U9: sources/stacks-chow/tags.txt, source lines 2402–2402. Navigation pointer for \ref{morphisms-lemma-base-change-flat} used inside the counted span (line 2425); the argument that consumes it is retained in the span.

Source reference. Original lemma 01W4: sources/stacks-chow/tags.txt, source lines 2469–2469. Navigation pointer for \ref{morphisms-lemma-base-change-proper} used inside the counted span (line 2422); the argument that consumes it is retained in the span.

Source reference. Original lemma 02NK: sources/stacks-chow/tags.txt, source lines 3483–3483. Navigation pointer for \ref{morphisms-lemma-base-change-relative-dimension-d} used inside the counted span (line 2424); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 2402); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y \end{array}$$

be a fibre product diagram of schemes locally of finite type over S. Assume $f : X \to Y$ proper and $g : Y' \to Y$ flat of relative dimension r. Then also f' is proper and g' is flat of relative dimension r. For any k-cycle α on X we have

$$g^* f_* \alpha = f'_* (g')^* \alpha$$

in $Z_{k+r}(Y')$.

Proof. The assertion that f' is proper follows from Morphisms, Lemma 01W4. The assertion that g' is flat of relative dimension r follows from Morphisms, Lemmas 02NK and 01U9. It suffices to prove the equality of cycles when $\alpha = [W]$ for some integral closed subscheme $W \subset X$ of δ-dimension k. Note that in this case we have $\alpha = [\mathcal{O}_W]_k$, see Lemma 02QY. By Lemmas 02R6 and 02RE it therefore suffices to show that $f'_*(g')^* \mathcal{O}_W$ is isomorphic to $g^* f_* \mathcal{O}_W$. This follows from cohomology and base change, see Cohomology of Schemes, Lemma 02KH. □